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COSC2753|COSC2812 - MACHINE LEARNING

ASSESSMENT TASK 2: DESIGN DEVELOPMENT AND APPLICATION

FASHION INTELLIGENCE SYSTEM

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"I declare that in submitting all work for this assessment. I have read, understood, and agree to the content and expectations of the Assessment declaration."

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1. Problem Framing and Evaluation Design

One catalogue of 38,612 usable low-resolution product photographs [3] supports four tasks, all trained from scratch: 92-class item type, season, joint gender and usage, and top-K visual retrieval. Seller labels partly record how an item was marketed rather than what is visible, which caps what any image model can recover. Our central argument, tested rather than assumed, is that the training distribution and the label convention bound performance more tightly than capacity or resolution do.

Metrics and splits. Class distributions are uneven in every task, so accuracy is never quoted alone. Task 1 selects on weighted-F1 and diagnoses on macro-F1, because across three seeds weighted-F1 holds at 80.48 with a spread of 0.24 while macro-F1 spreads by 0.75. Tasks 2 and 3 lead on macro-F1, Task 3 adding exact match across both heads; Task 4 reports P@10 paired with colour@10 and both@10 against a 5.91% random floor. Tasks 1 and 3 share one 71.4/14.3/14.3 partition grouped on a key that merges product names with byte-identical image families and stratified on the target; Tasks 2 and 4 group on product name alone. Near-identical but byte-distinct variants survive both guards, so we call the splits duplicate-free, not leakage-free.

Discipline. Selection happens on validation, any difference inside a stated noise floor is a tie, and every read of a test partition is recorded. Two evaluation scripts measure Task 3 on slightly different rows (the same base weights read 90.33% gender accuracy in one and 89.78% in the other), so every before-and-after comparison uses the same script. Deployment is judged separately from accuracy: the four models serve one endpoint that accepts an uploaded photograph, so every deployed model is measured on held-out catalogue tiles and on the same garments composited onto scene photographs from categories no model saw in training.

External data. Two public sources are used and neither contributes a training label to Tasks 1 to 4: Places365 [2] supplies photographic backdrops for the robustness work, and the Street Fashion Style corpus [1] supplies the auxiliary layers of Section 3.5. Both are downloadable from the cited records, and the loaders that consume them are in the submitted repository.

2. Data: Audit, Decisions, and Preprocessing

Characteristics. 38,617 metadata rows across 12 columns and 38,612 usable images; the graded set is 5,829 unlabelled tiles. Raw articleType spans 125 classes at a 6,781:1 imbalance ratio, and usage holds 8 classes with Casual taking 76.90%. Majority-class baselines are 17.56% for articleType, 49.58% season, 54.17% gender and 76.90% usage, which is what justifies macro-F1 alongside accuracy ([Figure A1](#)).

Target retention. In the raw metadata 33 classes have 10 rows or fewer, seven of them a single row; one of these (Suits) has no image file and is removed by the validity check instead. On the 38,612 usable rows the cutoff removes 32 of 124 classes and 121 rows (0.31%), leaving 92. For usage, Smart Casual (55) and Travel (25) merge into Casual, Party (13) into Formal, and Home (1) is dropped with 72 unlabelled rows: four classes for 73 rows (0.19%). Pooling the rare usage classes into an Other class instead was measured and rejected, scoring F1 = 0.000 on that class and costing roughly 15 points of usage macro-F1.

Repairs, inconsistency and validity. Unquoted commas inside product names spill 21 rows into two phantom columns; rejoining before dropping recovers all 21, dropping first truncates them silently. No ids or full rows are duplicated, but 8,972 rows match once id is ignored and 15,513 share a product name, and those groups carry conflicting season, usage and articleType labels in 907, 282 and 109 cases. Season tracks catalogue year rather than appearance: 88.2% of 2015 items are Winter. The 1,734 year values outside the IQR bounds are kept, since the range is 2007 to 2018 and none is implausible. Of the 38,612 images, 17 are not at the modal 60x80 resolution and 343 (0.89%) are greyscale; every loader converts to RGB and resizes, verified on those atypical cases rather than the easy ones, because a batch loader fails on frames of different shapes.

Split integrity. Hashing at full resolution finds 636 byte-identical image families across 1,399 files. As [Figure A4](#) shows, 218 span different product names, leaving 470 rows exposed to cross-split duplication under name-only grouping, and naive stratification leaks 1,657 names outright. The split_group key therefore combines names with hash families. Excluding 41 further rows whose duplicates carry conflicting gender or usage labels leaves 38,571 supervised items.

The domain gap, the largest risk in this project. Sampled channel means are [0.853, 0.833, 0.827] at an overall brightness of 0.836: the catalogue is a near-uniform white studio background, centred and evenly lit. A model fitted to frames this bright has no reason to be invariant to a photograph that has none, so Section 3 treats the training backdrop as an experimental variable rather than a fixed property of the data. Tasks 2 and 3 run at 60x80 and Tasks 1 and 4 at 120x160, with resolution likewise measured rather than assumed.

3. Algorithms Considered

3.1 Task 1: Item type

The ladder. On one split and one preprocessing contract, a classical HSV and gradient descriptor with logistic regression reaches 56.15 validation weighted-F1, a pixel MLP 67.81, and a CNN 79.04, rising to 80.14 when one vertical pooling division is kept, against a 5.28 majority floor (Table A1). The descriptor records how much structure is present but not where it sits, which is what convolution encodes, so the CNN is adopted.

Tuning. Balanced class weighting cost 15.32 weighted-F1 (82.20 to 66.88) and was rejected. Training to the full epoch budget added 5.84 (82.20 to 88.04), more than any architectural or inference-time change tried. Horizontal-flip test-time augmentation moved validation +0.28 weighted-F1 against a seed spread of 0.24, a tie, and is deployed only because it is nearly free; logit adjustment, tuned on validation to lift macro-F1, moved test macro-F1 by -0.36 and was declined. The 60x80 development network has 428,620 parameters and scores 87.64% test accuracy and 87.13 weighted-F1 with no test-time augmentation applied. A frozen ImageNet ResNet-18 with a logistic probe reaches 76.40 on the same split: a reference only, not a submitted model, and a floor rather than a verdict, since 32x downsampling leaves a 60x80 input only a 2x3 feature map.

Cost of the cutoff. The graded set does not share the training class prior: it is 21% Personal Care against 2.6% of training rows, so the seven dropped Personal Care classes, 26 training rows, account for 31.8 of an estimated 49.9 graded rows that belong to a class the model cannot output, against 18 under the flat training prior; the graded figure is the one that matters. A cutoff of 5 gives 103 classes but 54.73 weighted-F1 against 56.15 on the logistic proxy, so the threshold stays at 10.

The 120x160 model. Retrained with AdamW, OneCycleLR, 40 epochs, flips and label smoothing 0.05, it reaches 90.23% test accuracy, 89.68 weighted-F1, 73.11 macro-F1 and 72.84 balanced accuracy on the shared partition. On rows that neither checkpoint trained on, it scores +0.39 weighted-F1 (interval [-0.55, +1.35]) and 0.9 lower on macro-F1, a tie in both directions, and upscaling the 60x80 graded tiles to its 120x160 input costs a further 3.97. Four times the pixels moved nothing the noise floor can see.

Background adaptation. Eight epochs of fine-tuning on a 70% Places365 and 30% procedural backdrop bank move clean weighted-F1 to 87.14 and held-out Places365 weighted-F1 from 16.84 to 57.96. The clean cost is larger than the weighted-F1 line suggests and both halves must be quoted: macro-F1 falls 73.11 to 63.88 and balanced accuracy 72.84 to 62.59, and on the graded set the adapted model predicts 66 distinct classes against 76. The loss sits in the rare tail, already this task's weakest part. Section 5 makes the deployment decision.

Error structure. Measured on the catalogue-only 120x160 reference checkpoint, family accuracy at subCategory level is 97.2% against 90.2% at articleType, and 71.7% of the error mass is symmetric confusion between adjacent labels: Casual Shoes against Sports Shoes (73 errors), Tops against Tshirts (42), Flats against Heels (42). Two-way confusion is two labels competing for one appearance. Pairwise specialists test that directly and lose: a binary CNN per pair reaches 87.71% and 93.49% on its whole pair population, but on the gated rows where the main model is unsure it fixes 43 and breaks 55. The model identifies the object; what remains is the name the catalogue chose.

3.2 Task 2: Season Classification

Data and architecture. 38,592 usable observations remain after 20 missing labels, split 27,565 / 5,513 / 5,514 with zero group overlap. Augmentation is limited to horizontal flips, 5-degree affine jitter and 10% contrast jitter; hue jitter is excluded on principle, since a strong hue shift turns a Summer palette into an Autumn one and corrupts the label it is meant to preserve. Three paired 3x3 convolution blocks with batch normalisation and global average pooling, 304,932 parameters, Adam, early stopping at epoch 27 of 30. Global pooling rather than a flattened head keeps capacity proportionate to a four-class target whose rarest class holds 1,566 rows.

Imbalance and result. Inverse-frequency weights are applied in the loss (Spring 6.158, Summer 0.504). Task 1 measured the same correction costing 15.32 weighted-F1 and rejected it, so the choice is made per task rather than globally: defensible at four classes and 12:1, where it was not at 92 classes and 6,781:1. Test accuracy is 67.54% with 64.32 macro-F1 and 69.91 balanced accuracy, against a majority floor of 49.58% and 16.57. Balanced accuracy exceeding accuracy, with Spring recall at 73.5% against 44.3% precision, confirms the weighting bought rare-class recall and paid for it in precision ([Figure A2](#)).

What this task does not establish. One architecture was trained once here. There are no seed repeats, so Task 2 has no noise floor; the weighting was adopted on Task 1's evidence rather than ablated on this split; and no comparative claim is made beyond the majority baseline. We state this rather than let a single number read as a selected one. Of 5,514 test items 1,790 are misclassified (32.46%), and the most confident errors are watches, sunglasses and jewellery

predicted at up to 0.999 confidence, items whose photographs carry no seasonal evidence at all. With 88.2% of 2015 items carrying a Winter label, the ceiling is in the label rather than the classifier.

3.3 Task 3: Gender and Usage

Architecture. One network with early branching: two shared convolution blocks, then a separate two-block pathway and head per attribute, so the attributes share low-level evidence and separate where they diverge. The trunk holds 65,760 of 4,249,577 parameters. The 38,498 usable rows split 27,497 / 5,500 / 5,501 with no product and no byte-identical photograph in more than one partition. A constant predictor reaches 77.11% usage accuracy at 21.77 macro-F1, which is why macro-F1 leads.

Imbalance as an experiment. No correction, balanced sampling, balanced sampling with augmentation, and class-weighted loss were each trained across seeds 42, 7 and 13, and every figure here is the three-seed mean. The sampling configurations fail structurally: Ethnic is 95.5% Women and Formal 94.9% Men, so equalising usage through a shared trunk makes the rare genders rarer still, Girls falling from 1.6% to 0.6% of images seen and gender macro-F1 by 0.76 without augmentation and 2.15 with it. The weighted loss ties at -0.26 on mean macro-F1, its recall gains on Unisex, Sports and Girls paid for in precision on the same classes.

Selection. Three configurations sit inside the +/-0.67 threshold set by validation sampling error, wider than the +/-0.46 seed variance, so the rule falls through to the simplest pipeline: no correction, seed 42 ([Figure A5](#)). An override to balanced sampling with augmentation was deployed for Sports recall, measured and reverted: Sports recall rose 64.8% to 83.4% but precision fell 77.7% to 63.3%, Sports F1 moved only 70.7 to 72.0, and gender macro-F1 fell 2.69. The test partition was therefore read twice, but the evidence that reversed the override is already present in validation, so no conclusion depends on the second read.

What is deployed. The selected configuration is then fine-tuned for eight epochs with photographic backdrops at learning rate $3e-4$. Within the adaptation harness, clean mean macro-F1 moves 78.54 to 79.95 and held-out Places365 mean macro-F1 25.74 to 55.24. The served checkpoint scores 90.20% gender accuracy and 77.23 gender macro-F1, 90.33% usage accuracy and 82.67 usage macro-F1, and 81.59% exact match. The inherited temperatures (5.83, 5.94) were fitted to weights that no longer exist and score worse than none on the adapted model, so they were refitted on clean validation to 2.139 and 1.999, cutting expected calibration error to 2.20 and 1.03; temperature is monotonic, so no prediction changes.

Ceilings. An oracle predicting usage from ground-truth `articleType` reaches only 54.87% Sports recall against the network's 64.78, because training files 4,169 Tshirts as Casual against 679 as Sports. On gender, 39% of the 532 errors in the selection harness place or displace Unisex, and forgiving those alone would lift gender accuracy from 90.3% to 94.1%; a gender word in the product name matches the label on 98.8% of test items that state one. Accessories scored 85.2% at 120x160 against 85.5% here, so the constraint is the label convention, not capacity or resolution.

3.4 Task 4: Visual search

Protocol and encoder. With no fixed label set to predict, the system learns a 128-dimensional embedding and ranks the 38,571-item gallery by cosine similarity, using `articleType` as the relevance proxy against a 5.91% random floor. Whole products are held out (4,129 of 27,527) and 2,000 unseen queries are sampled at most 40 per class; P@10 is always paired with `colour@10` and `both@10`, and `subCategory`, which no loss ever saw, is scored as a control. The paired sampling floor is +/-1.10. The encoder is four convolution blocks with a 128-d projection, 1,228,171 parameters, batch-hard triplet loss at margin 0.3 plus auxiliary `articleType` and `baseColour` heads dropped at inference. We measured and removed four earlier methods; the plain triplet's 92.33 P@10 was circular, and it ranked colour poorly, which is why the colour head exists. Against the hand-written 128-bin HSV histogram the encoder scores 76.19 P@10 and 55.23 `colour@10` to 28.67 and 47.68, and fusion and colour re-ranking land within 0.22 of `both@10` 41.98, ties, so the simplest pipeline ships.

The training distribution is the lever. [Figure A6](#) fixes architecture, recipe, seed and split and changes only the training backdrop. Arm C, trained on the catalogue as it ships, is the best model in the project on clean frames at 81.64 P@10 and near the worst on a photograph at 11.84. Arm P, procedural backdrops only, scores 60.13 on unseen patterns but 42.22 on unseen scenes, a 17.91-point gap which shows it learned to ignore procedural patterns, not backgrounds in general. Arm D, the deployed 70% Places365 and 30% procedural mix, scores 56.32 and 55.78, a 0.54-point gap. Paired on `both@10`, D loses 5.62 in domain (interval [-6.63, -4.65]) and gains 23.47 to 26.03 out of domain, with every interval clear of zero. Selecting on a clean headline would have shipped the model that fails in deployment. Resolution, by contrast, is a tie: a 60x80 arm of the same recipe differs from 120x160 by under a point on clean, hard and wild and by 1.66 and 2.22 on photo and wildphoto, for 4.3 times the training cost.

A cross-task idea, measured and rejected. Adding Task 1's class probabilities as a re-ranking bonus over the Task 4 shortlist (weight 0.10, tuned on 500 queries, scored on a disjoint 1,000) lifts clean P@10 from 73.01 to 84.20 and moves held-out-background P@10 from 57.34 to 57.17. The clean gain is an artefact of the metric: relevance means sharing the query's articleType, so the bonus promotes exactly what the metric counts as correct and measures Task 1's accuracy rather than Task 4's retrieval; out of domain, both models fail on the same background and cannot rescue each other. Not deployed.

3.5 Auxiliary Recommendation Layers

Task 2B and Task 3B. Two lightweight layers add advice the catalogue labels cannot give, both built on Street Fashion Style metadata [1] with GroupShuffleSplit separating users so repeated contributors cannot straddle splits. Task 2B predicts suitable season from 284,098 usable rows over 50 garment categories and 4,967 users: the selected class-balanced logistic regression reaches 35.82 validation macro-F1 against 34.07 unweighted, and on held-out users 41.74% accuracy and 36.89 macro-F1 against a majority baseline's 37.04% and 13.52. Task 1's top-1 articleType maps onto one of the 50 SFS categories, supporting 67 of 92 classes. Task 3B ranks occasions from 284,218 usable posts by category prior, lift against global frequency, a support factor and usage-compatibility weights. Both measure association in street-photo metadata, not end-to-end recommendation accuracy on an upload; the scores are heuristic rather than calibrated, they inherit errors from Tasks 1 and 3, and they leave every Task 1 to 4 prediction and metric unchanged.

4. Evaluation

In domain and out. Table A2 reports each deployed model on held-out catalogue tiles and on the same garments composited onto held-out Places365 scenes. Every model clears its naive floor by a wide margin, and season gains the least over its floor, consistent with a label only partly visible in the image. Task 3 is reported there in macro-F1 because its accuracy on backgrounds would still read 77% and 84% while Casual is 77% of usage: that is what a model degenerating toward the majority class looks like if accuracy is the only column.

Where the residual error sits. Inside the catalogue the ceiling is the label convention, and two tasks measured independently show the same shape: Task 1 scores 97.2% at family level against 90.2% at articleType with 71.7% symmetric error mass, and Task 3's oracle recovers less Sports recall from the true articleType than the network recovers from pixels. Neither is a modelling failure more capacity would fix.

The controls are what license the robustness claim. Each background-adapted arm in Tasks 1, 2 and 3 was run against a control given the identical eight extra epochs with no compositing (Table A3). Out of domain those controls moved by +0.09, -0.85 and +0.11 points: extra training on white tiles buys essentially no robustness, so the entire out-of-domain gain belongs to the backdrops. The clean side does not behave the same way, and the honest reading needs both halves: Task 3's clean control reaches 80.19 mean macro-F1 against the adapted arm's 79.95, so Task 3's apparent clean gain is the extra training, and the backdrops cost about 0.24 of it.

Composited evaluation is an upper bound, not an estimate. On the 23 of 31 real photographs that carry a catalogue articleType, Task 4's deployed encoder scores 16.52 P@10 where its composited benchmark says 55.78, and Task 1 scores 17.4% top-1 and 39% at family level against 88% on tiles. As [Figure A7](#) shows, a cutout pasted onto a scene is still centred, evenly lit and alone in the frame. The ordering of the Task 4 arms survives the move to real photographs (16.52 against the catalogue-only control's 12.17) but the margin does not, falling from 4.7x on composites to 1.36x, and at $n = 23$ the sampling error is roughly ± 10 points, so only the gap to 55.78 is safe to quote. Improving upload segmentation closed most of that margin by itself, which says segmentation and background augmentation are substitutes: both remove the background, so their gains do not add.

Negative results, kept rather than deleted. Every idea declined in Section 3 is reported with the measurement that declined it, and resolution belongs on that list: at 120x160 Task 3 tied within ± 0.2 , Task 1 gained +0.39 with an interval straddling zero, and Task 4 tied on three of five benchmarks at 4.3 times the cost. Changing what the training images contain moved every task by tens of points; changing how many pixels they contain moved none past its floor.

5. Ultimate Judgement and Deployment

The judgement. The system is fit to deploy against catalogue-style product photography and is not fit for unconstrained user photographs without an explicit confidence gate. The binding constraint is not model capacity. It is the distance between the flat-lay images the models were trained on and the images a user submits.

Shipping is a separate question from work, and the four tasks answer it differently from the same evidence. Task 3 ships its adapted checkpoint because no axis loses against the incumbent: +1.41 clean and +29.50 out of domain within

one harness. Task 1 ships its adapted checkpoint at a real and stated cost: -2.53 weighted-F1, -9.23 macro-F1 and -10.25 balanced accuracy for +41.11 out of domain, roughly 16:1, the loss concentrated in the rare tail. We take that trade because the same checkpoint serves uploads, and because the catalogue-only model does not merely degrade out of domain, it collapses to 16.84. Task 4 ships arm D: 5.45 clean P@10 for +43.94 on photographs, roughly 8:1. Task 2 does not ship its arm: -3.50 for +8.99 is the weakest trade in the project, on the weakest model, for an attribute only partly present in the image. The intervention is identical in all four tasks; the decision is not, and that is the finding rather than an inconsistency. Promoting the Task 1 and Task 3 checkpoints changed 24.94% of graded articleType rows, 8.47% of gender and 4.77% of usage, season byte-identical.

The application implements that policy. A FastAPI service answers all four tasks from one upload, with per-task health reporting so one failed checkpoint degrades the system rather than downing it (Figure A3). An ingestion router sends catalogue tiles through a plain letterbox and uploads through segmentation, which alone lifted Task 3's gender accuracy under mild corruption from 36.9% to 65.9%, and a test walks all 5,829 graded tiles to assert that every one takes the catalogue branch. Retrieval returns a confidence verdict and answers 70.2% of test images confidently, because a system that can decline is more usable than one that fails silently.

A note on pretrained components. Every model that produces a prediction here is trained from scratch on the supplied data: the four task models and the retrieval encoder. One dependency deserves stating plainly. Upload ingestion segments the subject with a tiered ladder whose top tier is rembg (u2netp), a pretrained matting network. It is not used for training and not for any of the 5,829 graded predictions: Task 1 reaches them through a plain resize that never calls the segmenter, and Tasks 2 and 3 through the router above, which sends all 5,829 down the non-segmenting branch. It is used only for the real-photograph tables in Sections 3.4 and 4, which is why those are reported as end-to-end pipeline scores rather than properties of the learned models.

6. Limitations and Conclusion

Long tail. Dropping the 32 classes with ten or fewer examples leaves about 50 graded items that the model can never predict, and the deployed model's macro-F1 of 63.88 against a weighted-F1 of 87.14 shows that the promotion bought its robustness at the expense of the rare classes.

Season ambiguity. Season gains the least over its baseline because its labels follow the catalogue year rather than the garment's appearance, so its output should be read as a suggestion rather than a reliable attribute.

Multi-garment inputs. The encoder produces one vector per image, and although region search softens this limitation, 8 of the 31 photographs were excluded from scoring because they do not show a single catalogue garment.

Residual leakage and test reuse: We removed exact duplicates, but similar or related items may still cross splits because Tasks 2 and 4 group only by product name. Task 2 uses one run, and Task 3's test set was consulted twice, so these results need cautious interpretation

Limited evaluation evidence: The real-photo test has 24 manually labelled images, with only 23 scorable cases. Each background arm uses one training seed, and rerunning changed P@10 by 0.94 points. Small score differences should therefore not be ranked.

Conclusion. The binding constraints on this system are the training distribution and the label conventions, since capacity and resolution were both measured and tied. The system works well on catalogue photography, whereas unconstrained uploads need screening and qualified results, and the figure to quote for them is the measured 16.52 rather than the composited 55.78.

Appendix

Table A1. Task 1 algorithm ladder on one split and one preprocessing contract. Weighted-F1 selects; the ResNet-18 row is a reference only and is not a submitted model.

Model	Split	Acc.	Macro-F1	Weighted-F1
Majority baseline	val	17.63	0.33	5.28
Classical HSV + gradient, logistic regression	val	52.36	37.40	56.15
Pixel MLP	val	69.24	39.59	67.81
CNN, global pooled head	val	79.08	55.35	79.04
CNN, one vertical pooling division kept	val	80.30	60.00	80.14
CNN, full epoch budget (selected 60x80)	val	88.45	72.70	88.04
Frozen ImageNet ResNet-18 probe (reference only)	test	75.59	58.40	76.40
Selected 60x80 CNN, no test-time augmentation	test	87.64	73.09	87.13

Table A2. Each deployed model on held-out catalogue tiles and on the same garments composited onto held-out Places365 scene categories. One metric per row; Task 3 is reported in macro-F1 because its usage accuracy on backgrounds still reads 84% while Casual is 77% of the class.

Task	Target	Metric	Naive floor	Catalogue	Places365
1	articleType (92)	weighted-F1	5.2	87.14	57.96
2	season (4)	macro-F1	16.6	63.06	31.26
3	gender (5)	macro-F1	14.1	77.23	46.36
3	usage (4)	macro-F1	21.8	82.67	64.12
4	same type in top-10	P@10	5.9	76.19	55.78

Task 2's row is its matched robustness baseline, which is the only measurement paired with the Places365 column; its official held-out headline on the Section 3.2 split is 64.32 macro-F1. Task 1 and Task 4 clean figures are the deployed background-adapted models, not the catalogue-only references.

Table A3. Background adaptation against a matched control given the identical eight extra epochs with no compositing. The control isolates what the backdrops contribute from what the extra training contributes.

Task (metric)	Baseline clean	Adapted clean	Control clean	Baseline OOD	Adapted OOD	Control OOD
1 (weighted-F1)	89.68	87.14	89.47	16.84	57.96	16.93
2 (macro-F1)	63.06	59.56	63.91	31.26	40.25	30.41
3 (mean macro-F1)	78.54	79.95	80.19	25.74	55.24	25.85

Out of domain the controls move by +0.09, -0.85 and +0.11 points, so the entire out-of-domain gain belongs to the backdrops. On the clean side, the controls match or beat the adapted arms, so Task 3's clean improvement is the extra training rather than the compositing.

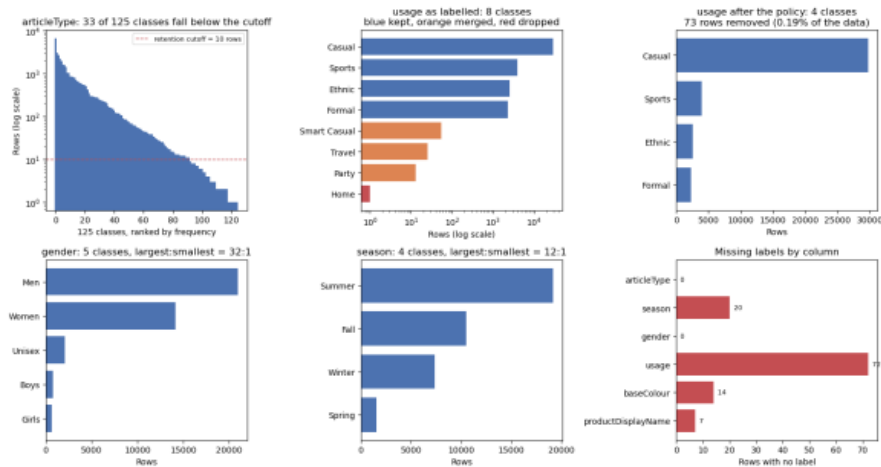


Figure A1. Target distributions and the usage label policy. articleType spans more than three orders of magnitude; blue is kept, orange merged, red dropped.

Classification report:

	precision	recall	f1-score	support
Fall	0.6708	0.6065	0.6371	1502
Spring	0.4432	0.7354	0.5531	223
Summer	0.7753	0.6639	0.7153	2734
Winter	0.5772	0.7905	0.6672	1055
accuracy			0.6754	5514
macro avg	0.6166	0.6991	0.6432	5514
weighted avg	0.6955	0.6754	0.6782	5514

Figure A2. Task 2 per-class season results on the held-out split. Spring recall 73.5% against 44.3% precision is the inverse-frequency weighting trading precision for recall.

Analyse an item

Drop in a garment photo. The four attribute models and the retrieval index all run against that one image.

dress.gif
648 x 960

Re-run Clear

No image handy?
Use a sample item

Predictions Inference: 6448.2 ms

Item type: T1
Dresses 32.1%
category: Dress 32.6%

Gender: T3
Women 99.3%

Season: T2
Fall 33.3%
Also suitable for: Spring / Summer

Occasion: T3
Casual 77.3%
Also suitable for: Dinner Date / Picnic

Confidence is the softmax probability of the winning class. Where a model is unsure, open alternatives to see what it was choosing between.

Similar items

Nearest neighbours by cosine similarity in the 128-dimension embedding space.

Whole image Per garment

6 12 24

#1 · id 43645
Dresses
Grey
78.2%

#2 · id 48488
Dresses
Black
75.5%

#3 · id 34566
Dresses
Blue
74.3%

#4 · id 51812
Nightdress
Blue
73.6%

#5 · id 51966
Nightdress
Blue
73.2%

#6 · id 49646
Dresses
Navy Blue
72.7%

Figure A3. The deployed application. One upload answers all four tasks, with calibrated confidence, the auxiliary season and occasion layers, and the retrieval confidence verdict.

One photograph, two product names: identical bytes that name-grouping splits



Figure A4. One photograph filed under two product names. 218 byte-identical families span different names, which is why name-only grouping is insufficient.

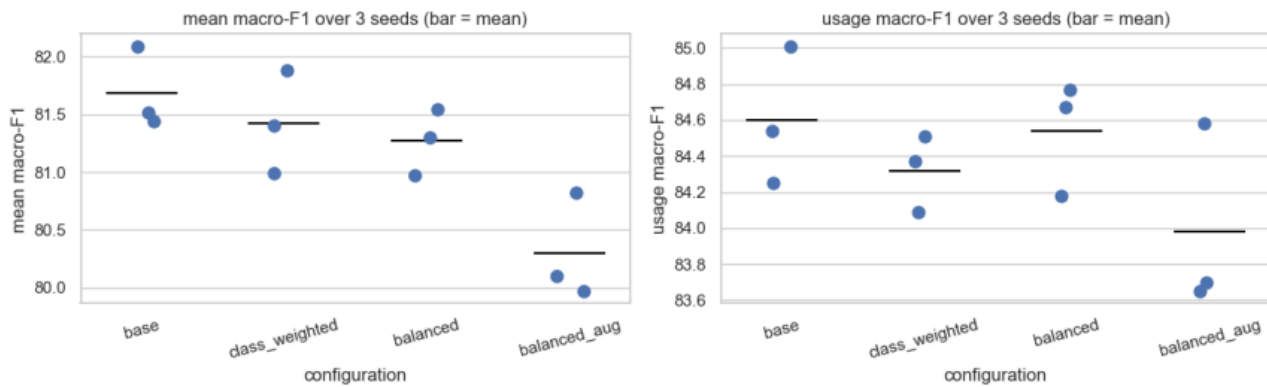


Figure A5. Task 3 configuration comparison across seeds 42, 7 and 13. Three configurations fall inside the ± 0.67 selection threshold, so the simplest pipeline is retained.

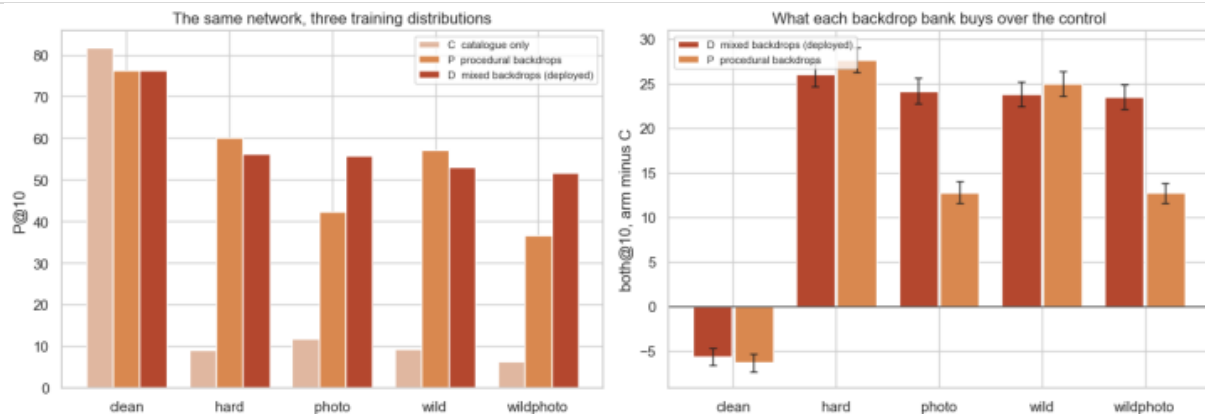


Figure A6. One network, three training distributions. Architecture, recipe, seed and split are fixed; only the training backdrop changes. Right: what each backdrop bank buys over the control on both@10.

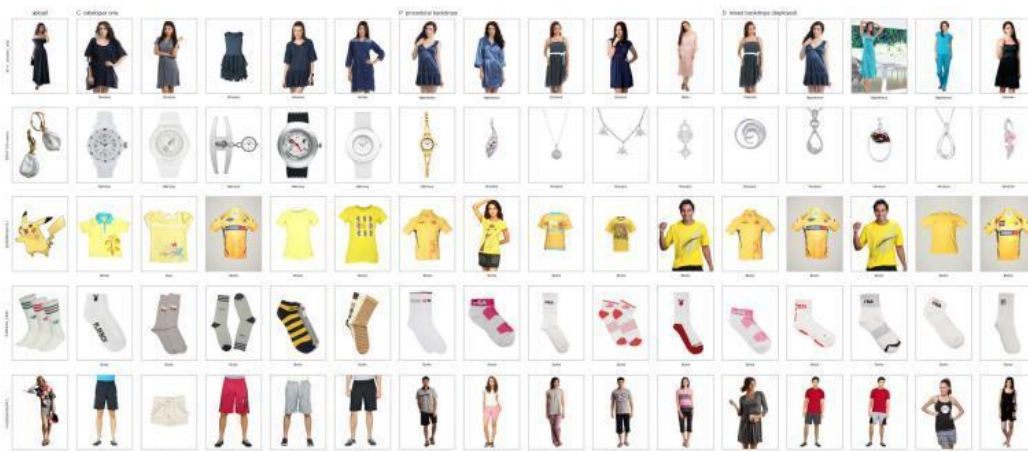


Figure A7. Five real uploads and their top five matches under each training distribution (C catalogue-only, P procedural, D deployed mix).

References

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